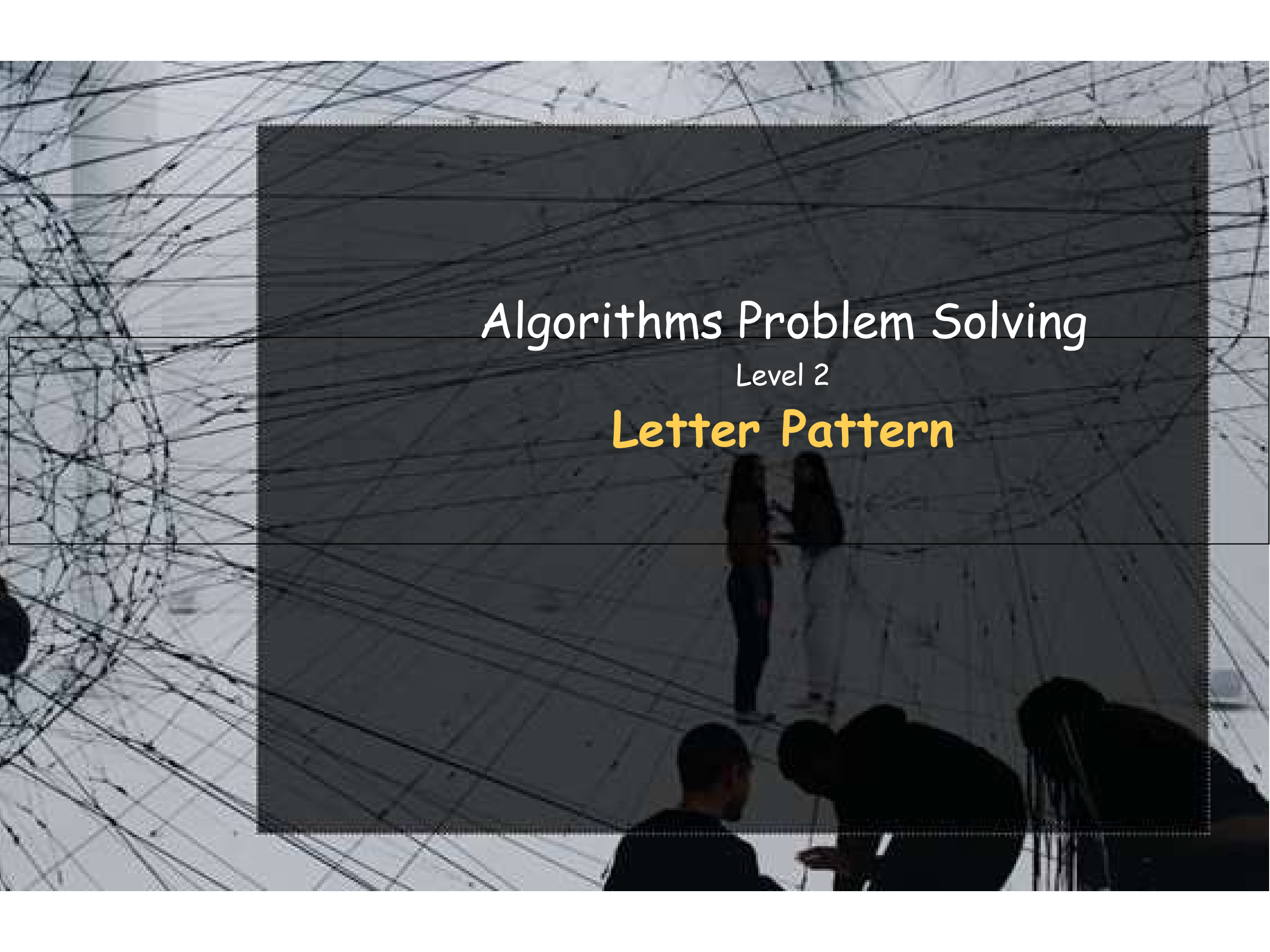


Algorithms Problem Solving

Level 2

Letter Pattern



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Letter Pattern

🔍 Definition

- ◆ A letter pattern is a pattern where letters are printed in ascending order, starting from `A`.
- ◆ In this program, the number of repetitions increases with each row.
- ◆ For example, input `3` produces `A`, `BB`, `CCC`.

Explanation

- ◆ ``ReadNumber()`` reads a non-negative number from the user.
- ◆ ``PrintLetterPattern()`` uses two nested ``for`` loops.
- ◆ ``65`` represents the character code of uppercase ``A``.
- ◆ The outer loop starts at ``65`` and continues up to ``65 + Number - 1``.
- ◆ ``char(i)`` converts the numeric character code into its corresponding letter.
- ◆ The inner loop controls how many times the current letter is printed.
- ◆ The number of repetitions increases from ``1`` to ``Number``.

Steps in main()

1. Read a non-negative number.
2. Start the outer loop from `'65' ('A')`.
3. Continue until `'65 + Number - 1'`.
4. Convert `'i'` to a character using `'char(i)'`.
5. Use the inner loop to print the current letter.
6. Move to the next line.
7. Increase the letter and the number of repetitions.

💡 Example

◆ Input:

`3`

◆ Starting character:

`65 → A`

◆ First row: `A` printed 1 time.

◆ Second row: `B` printed 2 times.

◆ Third row: `C` printed 3 times.

➡ Result

A

BB

CCC

Note

- ◆ The expression ``65 + Number - 1`` determines the last letter.
- ◆ For ``Number = 5``:
``65 + 5 - 1 = 69 → E``
- ◆ Therefore, the pattern starts at ``A`` and ends at ``E``.
- ◆ The inner loop uses the same calculation to determine how many times each letter is printed.

⊘ Warning

- ◆ The program allows `0` because the validation condition is $N < 0$.
- ◆ If the input is larger than the available uppercase English letters, the program can produce characters beyond `Z`.
- ◆ The code relies on the character encoding values for uppercase English letters.



Comparison

▨ #14_Inverted_Letter_Pattern

- ✓ Starts from the highest letter and moves backward to `A`.
- ✓ Repetitions decrease.
- ✓ Example: `3 → CCC, BB, A`.

▤ #15_Letter_Pattern

- ✓ Starts from `A` and moves forward.
- ✓ Repetitions increase.
- ✓ Example: `3 → A, BB, CCC`.

✦ Summary

- ✓ ``65`` represents uppercase ``A``.
- ✓ The outer loop moves from ``A`` upward.
- ✓ ``char(i)`` converts the character code into a letter.
- ✓ The inner loop controls the number of repetitions.
- ✓ Each row contains one more letter than the previous row.
- ✓ Input ``5`` produces ``A``, ``BB``, ``CCC``, ``DDDD``, ``EEEEEE``.

Thank you 😊